W-H-205C
24 May 1985
SUPERSEDING
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February 5, 1980

FEDERAL SPECIFICATION

# HEATING ELEMENTS, ELECTRICAL (REPLACEMENTS FOR DOMESTIC APPLIANCES)

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies:

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers metal sheathed electrical heating elements used as replacement items for residential water heaters, ranges, dishwashers, dryers, and hotplates.

1.2 Classification.

1.2.1 Types and styles. Heating elements shall be of the following types and styles, as specified (see 6.2).

Type I - Immersion units

Style A - Flange mounted
Style B - Screw-plug mounted

Type II - Nonimmersion units

Style A - Coiled surface units
Style B - Center mounted loop
Style C - U-shaped loop

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standards:

FED-STD-123 - Marking for Shipment (Civil Agencies)
FED-STD-H28 - Screw-Thread Standards for Federal Services

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards, and



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Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specification documents, and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specification:

MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of military specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the acquiring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Underwriters Laboratories Inc. (UL)

UL 1030 - Sheathed Heating Elements

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.



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## 3. REQUIREMENTS

3.1 Standard commercial product. The heating elements shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the heating elements being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.2 First article. When specified (see 6.2), the contractor shall furnish one heating element of the classification specified for first article inspection and approval (see 4.2.1, 6.2, and 6.4).

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts. All threaded parts shall conform to FED-STD-H28.

3.5 Codes and standards. The heating elements shall conform to the requirements of UL 1030 as applicable.

3.5.1 Compliance. Prior to approval of the first shipment, the contractor shall submit for the approval of the contracting officer, or his authorized representative, satisfactory evidence that the heating element he proposes to furnish under this specification meets the requirements of UL 1030 as applicable.

3.5.2 UL 1030. Acceptable evidence of meeting the requirements of UL 1030 shall be the UL certification symbol or label, listing in the UL files, or a certified test report (see 6.3) from a recognized independent testing laboratory indicating the heating element has been tested and conforms to UL 1030. Such evidence must be acceptable to the contracting officer.



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3.6 Electrical characteristics. The heating element shall be designed to operate on 60-Hertz alternating current at the voltage and wattage specified (see 6.2). Voltage shall be as follows:

| Voltage | Phase |
|---|---|
| 120 volts(V) | Single |
| 120/240V | Single |
| 240V | Single |
| 208V | Single/3 phase |

3.7 Construction. The heating element shall consist of straight or helically coiled nickel-chromium, or iron-chromium-aluminum-cobalt alloy wire mounted centrally within the tubular or cast metal sheath. Other resistance wire alloys of equivalent performance may be used when approved by the contracting officer or agency concerned. The space between the coil and sheath shall be solidly packed with magnesium oxide material, and in accordance with UL 1030. The ends of elements shall contain a moisture barrier to retard the entrance of water or vapors. Terminals shall be gas shield arc, projection or spot welded to the resistor element to ensure adequate strength and electrical conductivity.

3.7.1 Immersion elements. The sheath material shall be corrosion-resistant stainless steel or copper as specified (see 6.2). When specified (see 6.2), the copper shall be electroplated. The length of the heating portion of the element and whether it shall be a single or fold-back loop shall be as specified (see 6.2). A suitable flange gasket shall be provided for style A elements. The bolt pattern for style A elements shall be as specified (see 6.2). The National Pipe Thread size required for style B elements shall be as specified (see 6.2). The terminals of all immersion elements shall be moisture sealed. The temperature range of the heating element shall be as specified (see 6.2). When specified (see 6.2), a thermowell shall be a part of the immersion element. When specified (see 6.2), a thermostatic control shall be provided with each immersion element.

3.7.2 Nonimmersion elements. The element sheath shall be nickel alloy, corrosion-resistant steel tubing. Electrical connections shall be suitable for the application intended. When a particular type of electrical connection is required, it shall be as specified (see 6.2).

3.7.3 Coiled surface units (type II, style A). The style A sheath shall be formed with a flattened top surface to increase the conductive surface. The width of the flattened tube shall be at least 75 percent of the nominal tube diameter. A corrosion-resistant stainless steel supporting assembly, providing not less than three equally spaced support points shall be fastened to the coil by crimping or by use of retainer clips welded to the element. When specified (see 6.2), the nominal diameter of the element shall be 6 inches for an appliance opening of 6-1/2 to 8-1/4 inches in diameter or 8 inches for an appliance opening of 8-1/2 to 10-1/2 inches in diameter.

3.7.4 Center mounted (style B) and U-shaped (style C) loop elements. Elements shall be of the dimensions specified (see 6.2) and shall be compatible with the oven cavity. Unless otherwise specified (see 6.2), two supporting brackets shall be supplied with the baking element. Elements used for dishwashers shall be provided with a suitable water-tight sealing device.



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3.8 Grounding. Provision shall be made for grounding each heating element to the appliance. Grounding means shall be such that ground continuity will always be completed before power connections. The mounting screws of immersion heaters shall be considered an adequate ground.

3.9 Accessories. The following items shall be furnished as component parts when specified (see 6.2).

3.9.1 Adapter rings. Adapter rings for range and hotplate surface units shall be made of chrome-plated steel, or stainless steel, with sizes as specified (see 6.2).

3.9.2 Drip pans. Drip pans for range and hotplate surface units shall be made of chrome-plated, stainless, vitreous enameled steel, or aluminum, with sizes as specified (see 6.2).

3.9.3 Plug-in blocks. Range, bake and broil element terminal blocks shall be made of ceramic material with permanently mounted connecting lugs. Unless otherwise specified (see 6.2), the block shall accept elements with three prongs on 7/8-inch centers.

3.10 Identification markings. Identification markings shall be in accordance with UL 1030.

3.11 Workmanship.

3.11.1 Material fabrication. The material used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the material to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.11.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.11.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.11.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.



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3.11.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

3.12 Instructions. Each heating element ordered shall have installation instructions pertaining to that element. Safety instructions shall also be printed along with the installation instructions.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one heating element when a first article is required (see 3.2, 6.2, and 6.4). This inspection shall include the examination of 4.4, the tests of 4.5, and, when specified, the first article pack inspection of 4.6 (see 4.6 and 6.2). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5, and the preparation for delivery inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be one heating element. All elements offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be inspected using tightened inspection. If the rejected lot was screened, reinspection



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shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.3.1 Sampling for examination. Examination shall be based on inspection level II and an Acceptable Quality Level (AQL) of 1.5 for major defects and 4.0 for minor defects, expressed in defects per hundred units.

4.3.2 Sampling for tests. Tests shall be based on inspection level II and an AQL of 1.5 percent defective.

4.4 Examination. Each heater shall be examined for defects listed in table I. Each attribute within each classification of multiple defects shall constitute one defect.

TABLE I. Classification of defects.

| Classification | Defects | Requirement paragraph |
|---|---|---|
| Critical | | |
| None defined. | | |
| Major: | | |
| 101 | Materials not as specified, damaged, defective, or not suitable for the purpose intended. | 3.3 |
| 102 | Electrical requirements not as specified. | 3.6 |
| 103 | Configuration not as specified. Incorrect type, style or temperature range. | 3.7.1 |
| 104 | Electrical plugs, connections or terminals not as specified. | 3.7.2 |
| 105 | Provisions for grounding are inadequate. | 3.8 |
| 106 | Accessories: adapter rings, drip pans, plug-in blocks are incorrect or missing. | 3.9.1 thru 3.9.3 |
| 107 | Identification markings are missing, incorrect or illegible. | 3.10 |
| Minor: | | |
| 201 | Fasteners loose or missing. | 3.11.2 |



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4.4.1 Standards compliance. The contractor shall make available to the contracting officer or his authorized representative evidence of compliance with the applicable standard cited in 3.5. The Government reserves the right to examine and test all heater elements to determine the validity of the certification.

4.5 Tests. The first article and each production unit selected shall receive the tests of 4.5.1 and 4.5.2. Failure to pass any test shall constitute cause for rejection.

4.5.1 Dielectric voltage withstand test. The element shall be tested in accordance with the dielectric voltage withstand test of UL 1030.

4.5.2 Power input test. The power input test shall be in accordance with UL 1030 except that the power input shall not be more than 105 ercent nor less than 95 percent of its marked wattage or volt-ampere rating.

4.6 Preparation for delivery inspection. The inspection of the preservation, packaging, packing, and marking shall be in accordance with the requirements of section 4 of MIL-E-17555. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2), a first article pack shall be furnished for examination and test within the time frame required (see 6.2).

## 5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be in accordance with the requirements of MIL-E-17555 with the level of preservation and packaging and the level of packing as specified (see 6.2).

5.2 Marking.

5.2.1 Military agencies. Shipments to military agencies shall be marked in accordance with MIL-STD-129.

5.2.2 Civil agencies. Shipments to civil agencies shall be marked in accordance with FED-STD-123.

## 6. NOTES

6.1 Intended use. This specification covers immersion-type electrical heating elements for use in domestic hot water heaters; and nonimmersion tubular-type electrical heating elements for use in domestic ranges, hotplates, dishwashers, and dryers.

6.2 Ordering data. Purchasers shall select the preferred options permitted herein and include the following information in acquisition documents:

a. Title, number, and date of this specification
b. Type and style of element (see 1.2.1).
c. When a first article sample is required for inspection and approval (see 3.2, 4.2.1, and 6.4).



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d. Voltage and wattage required (see 3.6).
e. For immersion elements (see 3.7.1), specify:
   (1) Sheath material - stainless steel or copper.
   (2) When the copper shall be electroplated.
   (3) Length of loop.
   (4) Shape - single or folded loop.
   (5) Bolt pattern on style A elements.
   (6) Pipe thread size on style B elements.
   (7) Temperature range required.
   (8) When thermowell is required on style B elements.
   (9) When thermostatic control is required on style B elements.
f. For nonimmersion elements (see 3.7.2), specify:
   (1) When a particular type of electrical connection is required (see 3.7.2).
   (2) Surface units - for style A elements indicate diameter of the appliance opening and nominal diameter of the element (see 3.7.3).
   (3) For style B and C elements - indicate length and width of the oven cavity or give dimensions of the elements (see 3.7.4).
   (4) When supporting brackets are not required with baking element (see 3.7.4).
g. When adapter rings and drip pans for surface units are to be furnished as component parts, specify element diameter and appliance opening (see 3.9, 3.9.1 and 3.9.2).
h. When plug-in blocks are to be furnished as component parts (see 3.9 and 3.9.3). When block must accept other than three prongs on 7/8-inch center provide the following:
   (1) Spacing of prong holes.
   (2) Overall size.
   (3) Mounting requirements.
i. When a first article pack inspection is required and the time frame required for submission (see 4.2.1 and 4.6).
j. Level of preservation and packaging and level of packing required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of DoD Federal Acquisition Regulations (FAR) Supplement 27.410-6, the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DoD FAR 27.410-6 are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one heating element. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.



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MILITARY INTERESTS

Custodians:

Navy - YD

User Activity:

Navy - MC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

Navy - YD

DoD project 4540-0072



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